Zap Lab: Practice Test 1

Time: 50 minutes (15 stations, about 3 minutes each). Bring only a pencil; no notes or flashcards are allowed. You and your partner stay at each station until the coach calls time, then move to the next one; you may not skip ahead or go back. Stations 3, 13, and 14 have real low-voltage circuits on the table, and the rest are on this paper.

Partner names: _____ and _____ Score: _____

Station 1: Energy all around us

1. Which one is NOT a source of electrical energy?

◯ A battery ◯ B solar panel ◯ C light switch ◯ D generator at a power plant

2. A toaster changes electrical energy mostly into which kind of energy?

◯ A light ◯ B heat ◯ C sound ◯ D motion

3. Electricity is a form of _____ that can make devices work.

4. Name two things in your town or school (not at home) that use electricity, and what each one does.

Station 2: Conductor or insulator?

Materials list: copper wire, rubber boot, aluminum pie pan, wooden spoon, steel nail, glass window, plastic straw, silver ring.

5. Write every conductor from the list.

6. Write every insulator from the list.

7. Why do electricians use screwdrivers with thick rubber or plastic handles?

◯ A It makes the tool heavier. ◯ B The handle is an insulator, so electricity cannot reach the hand.
◯ C Rubber is a good conductor. ◯ D It makes the tool shiny.

8. In a wire, what job does the metal do, and what job does the plastic coating do?

Station 3: Test it

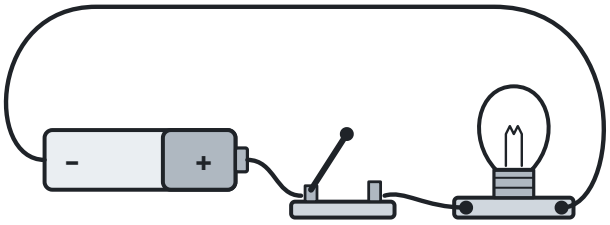
9. Use the tester on the table. Touch both clip ends to each object. Write C (conductor) or I (insulator) for objects 1, 2, 3, and 4.

10. What do the objects that made the bulb light have in common?

11. Why must the two clip ends NOT touch each other while you test an object?

Station 4: Parts of a circuit

This picture shows a battery, a knife switch, and a bulb joined with wires.



12. Name the power source in this circuit.

13. Name the load, and say what kind of energy it gives off.

14. What is the job of the switch?

◯ A It stores energy. ◯ B It makes the battery stronger. ◯ C It opens or closes the path for current.
◯ D It uses up the energy.

15. The switch lever is pointing up. What must you do to make the bulb light, and why does that work?

Station 5: Open or closed?

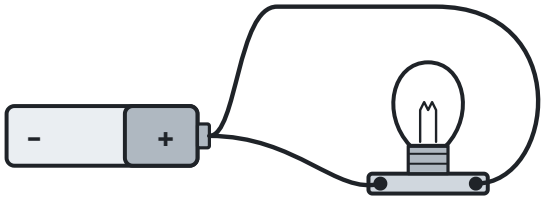
16. A circuit has a gap somewhere in its path. What kind of circuit is it?

◯ A closed circuit ◯ B open circuit ◯ C parallel circuit ◯ D short circuit

17. Jada's lamp is plugged in and has a good bulb, but its switch is set to OFF. Is the circuit open or closed? Why is the bulb dark?

18. A doorbell button is a kind of switch. When you press it, is the circuit open or closed, and what happens?

Station 6: One wire is in the wrong place



19. Both wires from the bulb holder go to the same end of the battery. Which end, + or -?

20. Why can no current flow through the bulb?

21. Move just ONE wire end so the bulb lights. Which wire end would you move, and where?

Station 7: Circuit symbols

22. A circle with an X inside is the symbol for a:

◯ A motor ◯ B lamp (bulb) ◯ C meter ◯ D switch

23. A zigzag line is the symbol for a:

◯ A wire ◯ B battery ◯ C resistor ◯ D LED

24. A cell symbol has one long line and one short line. Which line is the + side?

25. Draw the symbol for an open switch and the symbol for a closed switch.

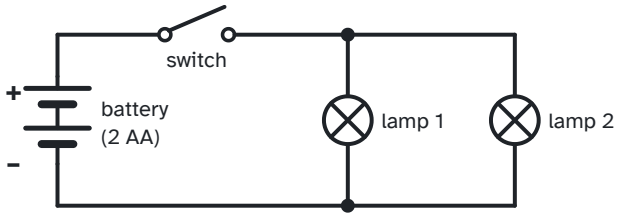
Station 8: Draw the schematic

26. Draw a schematic of a 2-cell battery, a closed switch, a motor, and a buzzer, all in one loop (series).

27. In your circuit, the motor breaks and leaves a gap. What happens to the buzzer? Why?

28. How many paths can the current take in your drawing?

Station 9: Read the schematic



29. Lamp 1 and lamp 2 are connected in:

◯ A series ◯ B parallel

30. With the switch closed, lamp 1 burns out. What happens to lamp 2? Why?

31. You want a switch that turns off ONLY lamp 2. Where would you put it?

Station 10: Series and parallel

32. Three bulbs are wired in series to one battery. Compared with one bulb alone on the same battery, each bulb is:

◯ A brighter ◯ B about the same ◯ C dimmer

33. Three bulbs are wired in parallel to the same battery. Compared with one bulb alone, each bulb is:

◯ A brighter ◯ B about the same ◯ C dimmer

34. An old string of holiday lights goes completely dark when one bulb is pulled out. Are the bulbs in series or parallel? How do you know?

35. Maya wants 2 lamps in her dollhouse. If one lamp burns out, the other must stay on. Should she use series or parallel? Draw a schematic.

Station 11: Current and voltage

36. Fill in the blanks: Current flows out of the ___ end of the battery, through the wires and the load, and back into the ___ end.

37. Adam adds a second AA cell in series with the first, and his bulb gets brighter. Explain why, using the word voltage.

38. Three AA cells (1.5 V each) are in series. What is the total voltage?

39. Adding more resistance to a circuit makes the current:

◯ A bigger ◯ B smaller ◯ C stay the same

Station 12: Units, resistors, and LEDs

40. Write the unit for each: voltage _____, current _____, resistance _____, power _____.

41. A light bulb box says 9 W. What does the W stand for?

◯ A wires ◯ B watts, the power the bulb uses ◯ C volts ◯ D weight

42. An LED is a diode. What does a diode let current do?

43. Circuit X has a 100 ohm resistor. Circuit Y has a 10,000 ohm resistor. Same battery, same kind of LED. Which LED is brighter? Why?

Station 13: Read the meter

44. Touch the red probe to + and the black probe to -. Write the reading for P, Q, and R, with the unit.

45. Which battery should be replaced first? How do you know?

46. A needle meter is marked 0, 2, 4, 6, 8, and 10 V, with one small mark halfway between each pair of numbers. The needle points at the small mark between 6 and 8. What is the reading?

Station 14: Circuit detective

47. The bulb in the bag will not light. Check one thing at a time. What is wrong?

48. Fix it. What did you change, and what happened?

49. Why does one backwards cell keep the bulb dark?

50. Name two other problems that can keep a bulb from lighting.

Station 15: Safety

51. Which is the safest choice?

◯ A Pull a plug out by its cord. ◯ B Use a hair dryer next to a full sink.
◯ C Dry your hands before you plug in a lamp. ◯ D Poke a paper clip into an outlet.

52. Why is it safe to touch the ends of an AA battery, but never safe to touch the metal inside a wall outlet?

53. You see a lamp cord with cracked plastic and bare metal showing. What should you do?

54. What is a short circuit, and why is it dangerous?

25	Open: a line with a gap and a short line tilted up like a gate. Closed: the short line lies flat across the gap, making one straight path.
Station 8: Draw the schematic	
26	One loop (a rectangle) with no branches: battery symbol (long-short-long-short lines), closed switch, a circle with M (motor), and the buzzer symbol, all joined by straight lines.
27	The buzzer stops. In series there is only one path, so one gap stops the current everywhere.
28	One.
Station 9: Read the schematic	
29	B parallel (each lamp has its own branch)
30	Lamp 2 stays on, because its branch is still a complete loop through the battery.
31	In lamp 2's own branch (between the joining dot and lamp 2), not in the main wire.
Station 10: Series and parallel	
32	C dimmer
33	B about the same
34	Series. There is only one path, so one missing bulb breaks the loop for all of them.
35	Parallel. Drawing: a battery with two branches, one lamp on each branch (a switch in the main wire is OK).
Station 11: Current and voltage	
36	+ (positive) and - (negative)
37	Two cells in series give more voltage (about 3 V instead of 1.5 V). More voltage is a bigger push, so more current flows and the bulb is brighter.
38	4.5 V (1.5 + 1.5 + 1.5)
39	B smaller
Station 12: Units, resistors, and LEDs	
40	volts (V), amps or amperes (A), ohms (the Greek letter omega), watts (W)
41	B watts, the power the bulb uses
42	Flow only one way through it (a one-way path).
43	X. It has less resistance, so more current flows through the LED.
Station 13: Read the meter Coach: set out 1 digital multimeter set to DC volts, 20 V range (red probe in V, black probe in COM), and 3 batteries labeled P (a fresh AA), Q (a used or nearly dead AA), and R (a 9 V). Measure them yourself first and write the readings here: P ___ V, Q ___ V, R ___ V.	
44	About: P 1.5 to 1.6 V, Q lower (often 1.0 to 1.3 V), R about 9 V (8 to 9.6 V). Accept readings within 0.1 V of the coach's readings.
45	Q. It reads well below the 1.5 V a fresh AA gives, so it is running low.
46	7 V
Station 14: Circuit detective Coach: in a zip bag labeled Station 14, build one loop with clip leads: a 2-AA battery holder with ONE of the two AA cells turned backwards, a closed switch, and a bulb in its holder. Check that the bulb stays dark, and that it lights when the cell is turned around. Turn the cell backwards again before each pair.	
47	One of the two AA cells is in backwards.
48	Turned the backwards cell around so both point the same way (+ to -). The bulb lit.
49	The two cells push in opposite directions, so their pushes cancel out (about 0 V total) and almost no current flows.
50	Any two: a loose or missing wire (gap), an open switch, a dead battery, a burned-out bulb, an insulator in the path, both wires on one battery end.

Station 15: Safety	
51	C Dry your hands before you plug in a lamp.
52	An AA battery gives a small push (1.5 V). A wall outlet gives about 120 V, a push strong enough to hurt or kill you.
53	Do not touch or use it. Tell an adult, who can unplug it by the plug and replace the cord.
54	A path straight from + to - with no load. Too much current flows, so the wire and battery get hot and can burn you or start a fire.